

# ዘበኛ - Zebegna

## Gate Officer Management System

### Software Requirements Specification (SRS)

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**Prepared By:** *Alem Desta*

**University:** Mekelle University

**Organization:** INSA

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# 1. Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS) document provides a complete and detailed description of the Zebegna (ዘበኛ) Gate Officer Management System. It defines functional and non-functional requirements, system features, external interfaces, and design constraints. This document serves as the primary reference for development, testing, and stakeholder validation.

## 1.2 Intended Audience

| Audience                 | Usage                                                      |
|--------------------------|------------------------------------------------------------|
| Project Sponsor / Client | Validate that requirements align with business goals       |
| Development Team         | Reference for implementation and technical decisions       |
| QA Engineers             | Basis for test case derivation and acceptance testing      |
| System Administrators    | Understanding of deployment and configuration needs        |
| Gate Officers & Admins   | Understanding of system capabilities and expected behavior |

## 1.3 Intended Use

This document shall be used to:

- Guide development and ensure feature completeness
- Serve as a contractual agreement between stakeholders and the development team
- Provide a baseline for system testing and acceptance criteria
- Support future maintenance, enhancement, and onboarding of new team members

## 1.4 Product Scope

Zebegna is a full-stack, web-based device registry and exit-approval system designed for organizational gate security operations. The system enables:

- **Device Registration**- Gate officers register devices (laptops, phones, tablets, etc.) with owner information, serial numbers, and exit reasons
- **Exit Approval Workflow**-Officers approve or manually verify device exits at security checkpoints
- **User Management** -Administrators create and manage gate officer accounts
- **Activity Auditing** - Complete audit trail of all system actions
- **Role-Based Access Control**-Strict separation between administrative oversight and operational gate functions

The system does not include hardware integration (barcode scanners, RFID), mobile native applications, or multi-tenant support in its current scope.

## 1.5 Definitions and Acronyms

| Term                | Definition                                                                                        |
|---------------------|---------------------------------------------------------------------------------------------------|
| SRS                 | Software Requirements Specification                                                               |
| JWT                 | JSON Web Token — a compact, URL-safe means of representing claims for authentication              |
| RBAC                | Role-Based Access Control                                                                         |
| CRUD                | Create, Read, Update, Delete                                                                      |
| EARS                | Easy Approach to Requirements Syntax                                                              |
| BDD                 | Behavior-Driven Development                                                                       |
| FMEA                | Failure Mode and Effects Analysis                                                                 |
| API                 | Application Programming Interface                                                                 |
| ORM                 | Object-Relational Mapping                                                                         |
| Gate Officer        | Security personnel stationed at entry/exit checkpoints                                            |
| Admin               | System administrator responsible for user management and oversight                                |
| Device              | Any electronic equipment (laptop, phone, tablet, camera, USB drive, etc.) subject to exit control |
| Exit Approval       | The act of authorizing a registered device to leave the premises                                  |
| Manual Verification | An alternative verification process for devices that cannot be verified through standard approval |

## 1.6 Motivation

At **INSA**, the current device registration process is time-consuming and prone to errors:

- Devices registered at one gate often require verification at another, creating delays and extra work for gate officers.
- Gate officers currently maintain **two separate logs** for each visitor, manually cross-checking entries—a process that is repetitive and error-prone.
- Delays and human errors frustrate employees and officers alike, making the system inefficient and stressful.

Even in other organizations like **CBE**, manual spreadsheets are still in use. While these methods may “work” in limited scenarios, they are clearly **inadequate for modern organizational needs**.

These challenges motivated the development of **Zebegna**: a centralized gate officer management system designed to **streamline device registration, simplify exit approvals, reduce errors, and provide real-time visibility** across all gates. By replacing outdated manual processes with a **digital solution**, Zebegna supports efficiency, accountability, and aligns with the **Digital Ethiopia vision** of modern, paperless operations

## 2. Overall Description

### 2.1 User Needs

#### Gate Officers need to:

- **Quickly search** for devices by serial number, device name, or owner name
- **Register new devices** with complete owner information upon entry
- **Approve device exits** after verifying ownership and authorization
- **Perform manual verification** for devices requiring special handling
- **Edit or remove device records** when corrections are needed
- **View real-time device status** without manual page refresh

#### Administrators need to:

- Create and manage gate officer user accounts
- Enable/disable officer accounts without deletion
- Reset officer passwords when needed
- Monitor all system activity through a comprehensive audit log
- View the Gate Console in read-only mode for oversight purposes
- Ensure gate officers are the only personnel approving exits (separation of duties)

### 2.2 Assumptions and Dependencies

#### Assumptions

1. Users have access to a modern web browser (Chrome, Firefox, Edge, Safari — latest 2 versions)
2. The organization has a stable network connection between gate checkpoint terminals and the server
3. PostgreSQL database server is available and properly configured
4. Java 17+ runtime is available on the deployment server
5. Node.js 18+ is available for frontend build and development
6. Each gate officer has unique credentials provided by an administrator
7. Devices have unique, identifiable serial numbers

**Dependencies**

| <b>Component</b> | <b>Dependency</b>     | <b>Version</b> |
|------------------|-----------------------|----------------|
| Frontend         | Next.js (App Router)  | 16.x           |
| Frontend         | React                 | 19.x           |
| Frontend         | Axios                 | 1.x            |
| Backend          | Spring Boot           | 3.2.x          |
| Backend          | Spring Security + JWT | 6.x            |
| Backend          | Spring Data JPA       | 3.2.x          |
| Database         | PostgreSQL            | 14+            |
| Migration        | Flyway                | 9.x            |

## 3. System Features and Requirements

### 3.1 Functional Requirements

#### 3.1.1 Numbered and Described

| ID    | Requirement                                                                                                                   | Priority |
|-------|-------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-01 | The system shall allow users to authenticate using a username and password                                                    | High     |
| FR-02 | The system shall issue a JWT token upon successful authentication                                                             | High     |
| FR-03 | The system shall support two roles: ADMIN and GATE OFFICER                                                                    | High     |
| FR-04 | Gate officers shall be able to register new devices with device name, type, serial number, owner information, and exit reason | High     |
| FR-05 | Gate officers shall be able to search devices by serial number, device name, or owner name                                    | High     |
| FR-06 | The system shall provide serial number autocomplete suggestions after 2+ characters                                           | Medium   |
| FR-07 | Gate officers shall be able to approve device exits (changing status from PENDING to APPROVED)                                | High     |
| FR-08 | Gate officers shall be able to perform manual verification on devices                                                         | High     |
| FR-09 | Gate officers shall be able to edit pending device records                                                                    | Medium   |
| FR-10 | Gate officers shall be able to soft-delete device records with confirmation                                                   | Medium   |
| FR-11 | Admins shall be able to create new gate officer accounts                                                                      | High     |
| FR-12 | Admins shall be able to enable/disable officer accounts                                                                       | High     |
| FR-13 | Admins shall be able to reset officer passwords                                                                               | Medium   |
| FR-14 | Admins shall have read-only access to the Gate Console (no action buttons)                                                    | High     |
| FR-15 | The system shall log all actions (register, edit, delete, approve, manual verify) with user, timestamp, and details           | High     |
| FR-16 | Admins shall be able to view the complete activity log                                                                        | High     |
| FR-17 | The device list shall auto-refresh every 10 seconds                                                                           | Medium   |
| FR-18 | The device table shall support sorting by device name, serial number, owner, date, and status                                 | Low      |
| FR-19 | The system shall display toast notifications for all user actions                                                             | Low      |
| FR-20 | The system shall redirect unauthenticated users to the login page                                                             | High     |
| FR-21 | The default device type for new registrations shall be "Laptop"                                                               | Low      |

#### 3.1.2 EARS Format Requirements

| ID    | EARS Requirement                                                                                                                                                             |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-01 | When a user submits valid credentials on the login page, the system shall authenticate the user, issue a JWT token, and redirect to the appropriate dashboard based on role. |



|       |                                                                                                                                                                                                                |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-04 | When a gate officer submits the device registration form with all required fields, the system shall create a new device record with PENDING status and associate it with the specified or newly created owner. |
| FR-05 | When a user types in the search bar, the system shall filter the device list within 300ms to show matching results by serial number, device name, or owner name.                                               |
| FR-06 | When a user types 2 or more characters in the search bar, the system shall display up to 6 autocomplete suggestions from existing serial numbers.                                                              |
| FR-07 | When a gate officer clicks "Approve" on a PENDING device, the system shall change the device status to APPROVED, record the verifying officer, and log the action.                                             |
| FR-08 | When a gate officer clicks "Manual" on a PENDING device, the system shall change the device status to MANUAL, record the verifying officer, and log the action.                                                |
| FR-10 | When a gate officer confirms device deletion, the system shall perform a soft delete (marking is_deleted=true) and log the action.                                                                             |
| FR-14 | When an admin views the Gate Console with PENDING devices, the system shall display "Awaiting Officer" instead of action buttons.                                                                              |
| FR-17 | While the Gate Console is open, the system shall automatically refresh the device list every 10 seconds without user interaction.                                                                              |
| FR-20 | When an unauthenticated user attempts to access any protected route, the system shall redirect to the login page.                                                                                              |

### 3.1.3 BDD Specification (Gherkin Format)

#### Feature: User Authentication

##### Scenario: Successful login as gate officer

Given the user is on the login page

When they enter username "officer1" and password "officer123"

And they click "Log In"

Then they should be redirected to "/devices"

And a welcome toast should display "Welcome, Gate Officer One!"

And a JWT token should be stored in localStorage

##### Scenario: Failed login with incorrect credentials

Given the user is on the login page

When they enter username "admin" and password "wrongpass"

And they click "Log In"

Then an error message "Login failed. Check your credentials." should be displayed

And the user should remain on the login page

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#### Feature: Device Registration

**Scenario: Register a new device with a new owner**

Given the gate officer is logged in and on the Gate Console  
When they click "+ Register Device"  
And they fill in device name "MacBook Pro", type "Laptop", serial "SN-2024-001"  
And they fill in owner name "Abebe Kebede", type "EMPLOYEE", department "Engineering"  
And they select reason "Work"  
And they click "Register Device"  
Then the device should appear in the table with status "PENDING"  
And a success toast "Device registered" should appear

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**Feature: Exit Approval (Role Separation)****Scenario: Gate officer approves a device exit**

Given the gate officer is logged in  
And a device "SN-2024-001" exists with status "PENDING"  
When the officer clicks "Approve" on that device  
Then the device status should change to "APPROVED"  
And the "verifiedBy" field should show the officer's name  
And the action should be logged in the activity log

**Scenario: Admin cannot approve exits**

Given the admin is logged in  
And a device "SN-2024-001" exists with status "PENDING"  
Then the Actions column should show "Awaiting Officer" instead of approval buttons  
And no approve or manual verification buttons should be visible

---

**Feature: Activity Log****Scenario: Admin views activity history**

Given the admin is logged in  
When they click "Activity Log" in the sidebar  
Then they should see a table of all system actions  
And each entry should show timestamp, user, action type, and details  
And entries should be sorted by most recent first

---

## 3.2 Non-Functional Requirements

### 3.2.1 Performance

| ID      | Requirement                                                                        |
|---------|------------------------------------------------------------------------------------|
| NFR-P01 | 95% of API requests shall return a response within 2 seconds under normal load.    |
| NFR-P02 | The login page shall render within 1.5 seconds on a standard broadband connection. |
| NFR-P03 | Search results shall update within 500ms of the user finishing typing.             |
| NFR-P04 | The system shall support at least 50 concurrent users without degradation.         |
| NFR-P05 | The auto-refresh mechanism shall not block user interaction.                       |

### 3.2.2 Security

| ID      | Requirement                                                                    |
|---------|--------------------------------------------------------------------------------|
| NFR-S01 | Only authenticated users shall access any API endpoint except /api/auth/login. |
| NFR-S02 | Passwords shall be hashed using BCrypt before storage.                         |
| NFR-S03 | JWT tokens shall expire after 24 hours.                                        |
| NFR-S04 | Admin-only endpoints shall enforce ROLE_ADMIN authorization.                   |
| NFR-S05 | The system shall return HTTP 401 for expired or invalid tokens.                |
| NFR-S06 | CORS shall be restricted to explicitly configured origins.                     |
| NFR-S07 | CSRF protection is disabled due to stateless JWT architecture.                 |
| NFR-S08 | All activity shall be logged with the acting user's identity.                  |

### 3.2.3 Usability

| ID      | Requirement                                                                     |
|---------|---------------------------------------------------------------------------------|
| NFR-U01 | The UI shall be usable without training for basic operations (search, approve). |
| NFR-U02 | Toast notifications shall provide immediate feedback for all user actions.      |
| NFR-U03 | Form validation errors shall appear inline next to the relevant field.          |
| NFR-U04 | The system shall use a dark theme optimized for extended screen usage.          |

### 3.2.4 Reliability

| ID      | Requirement                                                                    |
|---------|--------------------------------------------------------------------------------|
| NFR-R01 | The system shall maintain data consistency through database transactions.      |
| NFR-R02 | Soft deletes shall be used to prevent accidental data loss.                    |
| NFR-R03 | Database migrations shall be managed by Flyway to ensure schema consistency.   |
| NFR-R04 | The system shall gracefully handle network errors with user-friendly messages. |

### 3.2.5 Compliance

| ID      | Requirement                                                         |
|---------|---------------------------------------------------------------------|
| NFR-C01 | The system shall maintain a complete audit trail of all operations. |
| NFR-C02 | User actions shall be attributable to specific authenticated users. |

## 3.3 External Interface Requirements

### 3.3.1 Performance Requirements

- Database: PostgreSQL 14+ with connection pooling via HikariCP (default pool size: 10)
- API Response Time: Average < 500ms, 95th percentile < 2s
- Frontend Bundle: Next.js optimized with route-based code splitting
- Auto-Refresh Interval: 10 seconds (configurable)

### 3.3.2 Safety Requirements

| ID     | Requirement                                                                          |
|--------|--------------------------------------------------------------------------------------|
| SAF-01 | Device deletion shall require explicit user confirmation via modal dialog.           |
| SAF-02 | Deleted devices shall be soft-deleted (recoverable) rather than permanently removed. |
| SAF-03 | Password fields shall support a show/hide toggle.                                    |
| SAF-04 | Session expiry shall redirect users to login without unintended data loss.           |

### 3.3.3 Security Requirements

#### Authentication Flow

User → Login Form → POST /api/auth/login → AuthService → BCrypt Verify → JWT Generate → Client Storage (localStorage)

#### API Security Matrix

| Endpoint                              | Method | Auth Required | Role Required |
|---------------------------------------|--------|---------------|---------------|
| /api/auth/login                       | POST   | No            | None          |
| /api/devices                          | GET    | Yes           | Any           |
| /api/devices                          | POST   | Yes           | GATE OFFICER  |
| /api/devices/{id}                     | PUT    | Yes           | GATE OFFICER  |
| /api/devices/{id}                     | DELETE | Yes           | GATE OFFICER  |
| /api/devices/{id}/verify              | POST   | Yes           | GATE OFFICER  |
| /api/devices/{id}/manual-verification | POST   | Yes           | GATE OFFICER  |
| /api/devices/suggestions              | GET    | Yes           | Any           |
| /api/activity-logs                    | GET    | Yes           | ADMIN         |
| /api/users/officers                   | GET    | Yes           | ADMIN         |

|                                         |      |     |       |
|-----------------------------------------|------|-----|-------|
| /api/users/officers                     | POST | Yes | ADMIN |
| /api/users/officers/{id}/reset-password | PUT  | Yes | ADMIN |
| /api/users/officers/{id}/toggle-status  | PUT  | Yes | ADMIN |

## 3.4 System Features

### Feature 1: Authentication and Authorization

- JWT-based stateless authentication
- Role-based access control (ADMIN, GATE\_OFFICER)
- Automatic token injection via Axios interceptors
- HTTP 401 handling with automatic redirect to login
- 24-hour token expiration policy

### Feature 2: Device Registry

- Full CRUD operations for device records
- Owner association (existing owner selection or new owner creation)
- Supported device types:
  - Laptop
  - Mobile Phone
  - Tablet
  - Camera
  - Hard Drive
  - USB Drive
  - Other
- Supported exit reasons:
  - Work
  - Reward
  - Other
- Optional supporting document reference field

- Default device type: Laptop

### Feature 3: Exit Verification Workflow

- Three-state lifecycle:
  - PENDING
  - APPROVED
  - MANUAL
- Only PENDING devices can transition to APPROVED or MANUAL
- Verification records include:
  - Verifying officer name
  - Timestamp
- Soft delete option for PENDING devices
- Real-time status updates via auto-refresh (10 seconds)

### Feature 4: Search and Discovery

- Full-text search across:
  - Serial number
  - Device name
  - Owner name
- Debounced search (300ms)
- Serial number autocomplete suggestions (minimum 2 characters, up to 6 suggestions)
- Status filtering
- Sortable table columns

### Feature 5: User Management (Admin Only)

- Create gate officer accounts
- Enable or disable accounts
- Reset passwords
- View all officers in a tabular interface
- Admin read-only access to Gate Console

## Feature 6: Activity Audit Log (Admin Only)

- Chronological record of all system actions
- Includes:
  - Timestamp
  - Acting user
  - Action type
  - Details
- Sorted by most recent first
- Pagination support (up to 500 entries per page)

## 4. Other Requirements

### 4.1 Database Requirements

**Database Engine:** PostgreSQL 14+

#### Entity-Relationship Overview

| Table        | Description                               |
|--------------|-------------------------------------------|
| users        | System users (admins and gate officers)   |
| persons      | Device owners (employees and guests)      |
| devices      | Registered devices and verification state |
| activity_log | Audit trail of all actions                |

#### Key Relationships

- devices.owner\_id → persons.id (Many-to-One)
- devices.registered\_by → users.id (Many-to-One)
- devices.verified\_by → users.id (Many-to-One, nullable)
- activity\_log.user\_id → users.id (Many-to-One)

#### Schema Management

- Flyway manages all database migrations in /db/migration/
- spring.jpa.hibernate.ddl-auto=validate enforces schema validation
- Default seed data includes initial admin and officer accounts

#### Default Seed Accounts

| Username | Password   | Role         |
|----------|------------|--------------|
| admin    | admin123   | ADMIN        |
| officer1 | officer123 | GATE OFFICER |

### 4.2 Legal and Regulatory Requirements

| ID    | Requirement                                                                  |
|-------|------------------------------------------------------------------------------|
| LR-01 | All user actions must be traceable to a specific authenticated user.         |
| LR-02 | Device exit records must be retained using soft deletion for audit purposes. |
| LR-03 | Passwords must be stored as cryptographic hashes and never in plain text.    |
| LR-04 | The system must comply with applicable organizational security policies.     |



### 4.3 Internationalization and Localization

| Aspect           | Current Support                                                         |
|------------------|-------------------------------------------------------------------------|
| Primary Language | English                                                                 |
| Secondary Script | Ethiopic (Ge'ez) — branding use ("ዘበኛ")                                 |
| Fonts            | Inter (Latin), Noto Sans Ethiopic                                       |
| Date Format      | MMM d, yyyy (e.g., Feb 25, 2026)                                        |
| Time Format      | 12-hour format with AM/PM                                               |
| Currency         | Not applicable                                                          |
| Future Support   | Architecture supports addition of Amharic translations via i18n library |

### 4.4 Risk Management (FMEA Matrix)

| Risk                          | Severity | Occurrence | Detection | RPN | Mitigation                                                     |
|-------------------------------|----------|------------|-----------|-----|----------------------------------------------------------------|
| Database server failure       | 9        | 2          | 3         | 54  | Automated backups; database replication                        |
| JWT token theft               | 8        | 3          | 5         | 120 | 24-hour expiry; HTTPS; future HttpOnly cookies                 |
| Unauthorized exit approval    | 9        | 2          | 2         | 36  | Strict RBAC; activity logging; admin oversight                 |
| Network outage at gate        | 7        | 4          | 2         | 56  | Future offline mode; local caching strategy                    |
| Incorrect device deletion     | 6        | 3          | 2         | 36  | Soft delete; confirmation modal; audit review                  |
| Password brute force          | 7        | 4          | 4         | 112 | BCrypt hashing; future rate limiting; account lockout          |
| Data inconsistency            | 8        | 2          | 3         | 48  | Transactions; Flyway migrations; schema validation             |
| Concurrent approval conflicts | 5        | 3          | 4         | 60  | Auto-refresh; future optimistic locking                        |
| Admin account compromise      | 10       | 1          | 3         | 30  | Change default credentials; strong password policy; future 2FA |

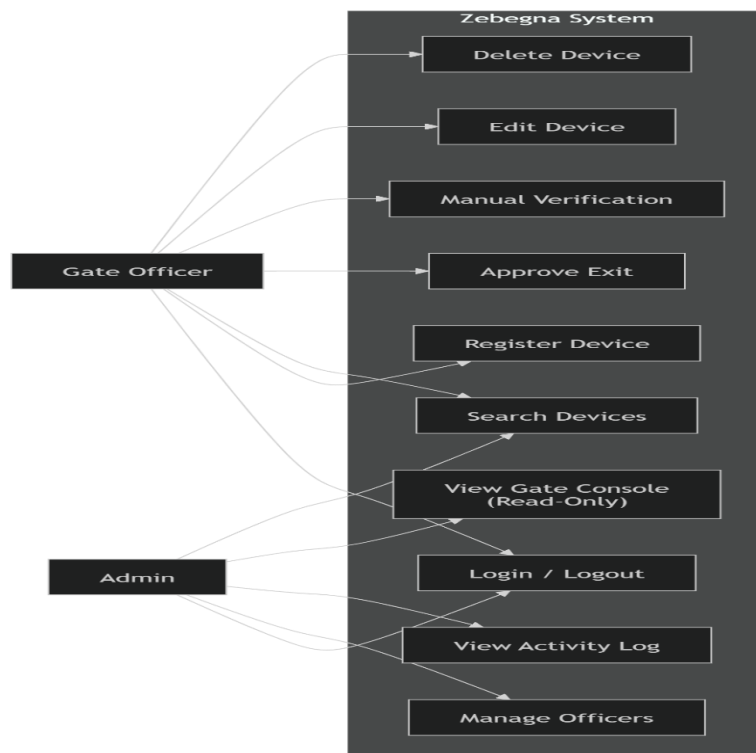
## 5. Appendices

### 5.1 Glossary

| Term                | Definition                                                                          |
|---------------------|-------------------------------------------------------------------------------------|
| Soft Delete         | Marking a record as deleted (is_deleted=true) without removing it from the database |
| Verification Status | Current state of a device: PENDING, APPROVED, or MANUAL                             |
| PENDING             | Device registered but not yet approved for exit                                     |
| APPROVED            | Device exit approved via standard verification                                      |
| MANUAL              | Device exit verified through manual process                                         |
| HikariCP            | High-performance JDBC connection pool                                               |
| Flyway              | Database migration and schema versioning tool                                       |
| Toast Notification  | Temporary UI message confirming an action                                           |
| Axios Interceptor   | Middleware that attaches JWT tokens to API requests                                 |

### 5.2 Use Cases and Diagrams

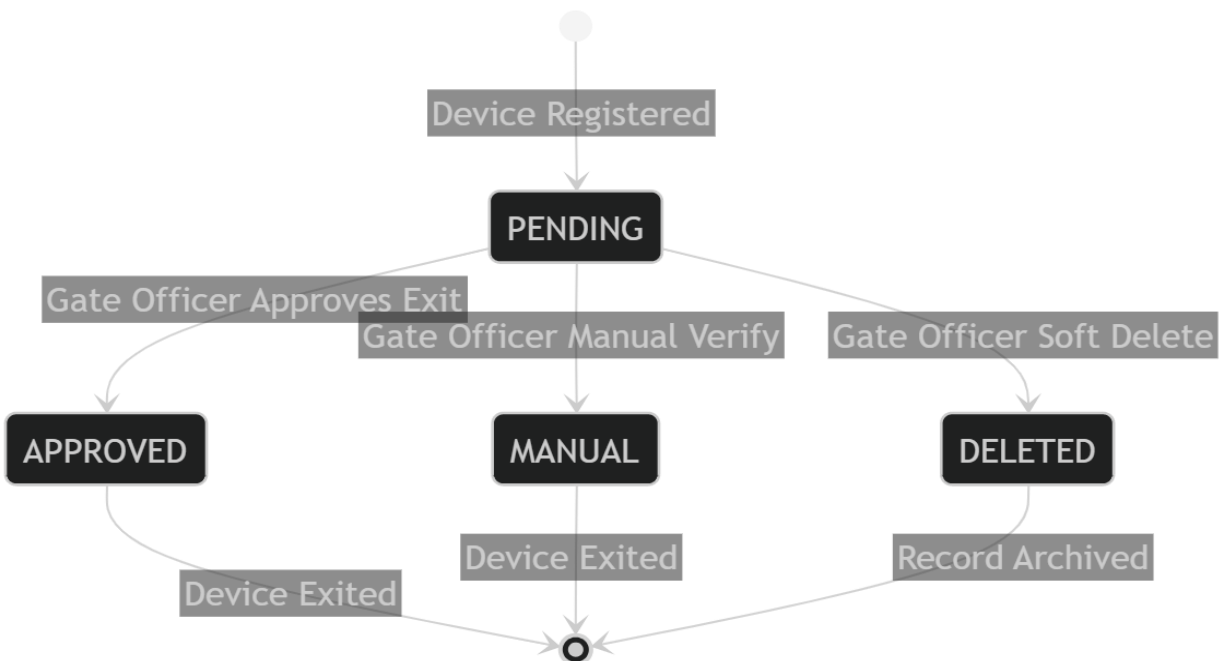
#### Use Case Diagram



## System Architecture



## Device Lifecycle State Diagram



### 5.3 To Be Determined (TBD) List

| ID     | Item                               | Priority | Notes                             |
|--------|------------------------------------|----------|-----------------------------------|
| TBD-01 | Offline mode for gate checkpoints  | Medium   | For unreliable networks           |
| TBD-02 | Multi-tenant support               | Low      | For multi-organization deployment |
| TBD-03 | Barcode/QR scanner integration     | Medium   | Hardware dependency               |
| TBD-04 | Email notifications for admins     | Low      | Unusual activity alerts           |
| TBD-05 | Two-factor authentication (2FA)    | High     | Admin security enhancement        |
| TBD-06 | Report export (PDF/CSV)            | Medium   | Compliance reporting              |
| TBD-07 | Device photo capture               | Low      | Visual verification aid           |
| TBD-08 | Login rate limiting                | High     | Brute-force mitigation            |
| TBD-09 | HTTPS/TLS production configuration | High     | Production security               |
| TBD-10 | Backup and disaster recovery plan  | High     | Data protection                   |
| TBD-11 | Light/Dark theme toggle            | Low      | User preference                   |
| TBD-12 | Amharic UI translation             | Medium   | Full localization                 |